

Primary LLM-DNA Robustness Experiment

Updated Full Experimental Protocol and Progress Report

Updated August 13, 2026

Executive summary

This document records the design and current execution status of the primary LLM-DNA decoding robustness experiment. The study will measure how temperature and top- p affect exact-model identification, whether a classifier can mitigate the resulting accuracy loss, and whether a multi-sample distributional representation improves robustness under a matched generation budget. The fixed primary cohort contains 100 validated models. The experiment contains one deterministic condition and a complete 4×3 stochastic grid, with two generation rounds per condition. The principal deployment setting assumes that temperature and top- p are hidden at inference time and only an upper bound on temperature is known. The frozen pre-recovery audit of August 5 contains 2,528 successful artifacts out of 2,600 planned slots. Of those successful artifacts, 572 record an explicit generation seed and 1,956 come from historical manifests that do not record a separate generation seed. Thus, 97.2% is an artifact-coverage statistic, not a seed-clean completion rate. The 72 unsuccessful slots affect 41 distinct models, leaving a strict all-grid common cohort of 59 in that snapshot. Targeted recovery was launched on August 13. Small-scale exploratory results are reported separately and are not substitutes for the final primary analysis.

Status and scope

Primary experiment status (pre-recovery snapshot, August 5, 2026): the grid contains 2,528/2,600 successful artifacts (97.2%). Two of 13 settings are fully complete. The 72 remaining slots are 52 failed_missing_file and 20 failed_empty_response; no ordinary pending slots remain. The strict intersection across all 26 runs contains 59 models and is below the preregistered minimum of 80. Section 12 contains explicitly labelled exploratory demonstrations that must not be interpreted as final primary findings. A failure-specific recovery job is running from the fixed 72-slot plan under ; all final counts must be regenerated after that job finishes.

A resource-feasibility review reduced the plan from 300 models and four repeats to 100 models and two rounds. Repeat 1 is the reference/training split and repeat 2 is held out. Exact identification is a 100-way closed-set task before failure filtering.

Frozen pre-recovery primary-grid snapshot from August 5. Each cell combines two 100-model rounds.

Temperature
0
0.2
0.3
0.5
0.7

The high slot-completion rate and the smaller complete-case cohort are not contradictory. Failures occur in different models across cells: 72 failed slots are distributed over 41 models, so $100 - 41 = 59$ models succeed everywhere. The 2,528 successful slots do include historical runs whose generation seed is unknown: 572 successes have an explicit generation seed and 1,956 do not. Unknown-seed runs remain usable for paired representation comparisons when every method reads the identical responses, but they are not treated as independent seeded evidence for causal decoding claims. For setting-specific paired evaluation, the usable r1–r2 cohorts remain much larger:

Paired cohort with both rounds successful in each setting.

T
0.0
0.2
0.2
0.2
0.3
0.7
0.7

The common cohorts for bounded analyses are 97 at $\tau=0.2$, 86 at $\tau=0.3$, 63 at $\tau=0.5$, and 59 at $\tau=0.7$. Targeted recovery should first repair 21 models that each lack only one slot; if successful, this is the minimum route to restore an 80-model all-grid cohort.

This experiment replaces the 77-model interim analysis as the primary robustness study. The earlier analysis remains a legacy diagnostic because it used an incomplete sweep, one generation repeat per reported cell, an historical reference gallery with unknown decoding parameters, and a binary same-or-related task that was not directly comparable to exact-model Top-1 retrieval.

Research questions and hypotheses

Primary research questions

1. How do temperature and top- p affect exact-model classification from LLM-DNA representations?
2. On the same exact-model task, can a classifier trained on the full DNA vector reduce the accuracy degradation relative to cosine retrieval?
3. Does the proposed multi-sample distributional representation improve robustness beyond original DNA under the same response and token budget?
4. If temperature and top- p are unknown at inference time, but temperature is known to satisfy $T \leq \tau$, what average and worst-setting accuracy can be expected?

Secondary research questions

1. When exact identification fails, does the predicted model remain close to the true model in the Hugging Face model tree?
2. Do direct parent-child, indirect, and disconnected model pairs exhibit different DNA-distance distributions as decoding stochasticity rises?
3. How sensitive are the conclusions to response encoder, truncation, projection dimension, projection seed, and the number of response samples?

The main hypothesis is that temperature will cause a larger reduction in exact classification accuracy than top- p over the planned range. Classifier and distributional-method benefits will be judged by paired accuracy differences and by a shallower degradation curve, rather than by comparing accuracies from different tasks.

Model cohort

Fixed validated cohort

The primary cohort is the 100-model list exported from the latest validated control manifest. Historical artifacts count only when their exact model ID, decoding cell, and round match this cohort and grid. Performance was not used to add or remove models.

The fixed artifacts are:

- cohort: ;
- canonical manifest: ;
- live audit: ; and
- cohort SHA-256:
9043a62b95436b9fc92671b6f5b11b15d427fddd745dba2f84c112d97e850e6d.

Any operational replacement would define a new cohort version and require the grid and every common-cohort count to be regenerated.

Completion requirement

The intended final analysis uses all 100 models. The automated primary analysis will refuse to run if fewer than 80 models are common to all required training and test runs. Missing models should first be backfilled. If the final common cohort remains below 100, the report will provide success rates and failure reasons by decoding cell and compare retained and excluded models by size and architecture.

Prompt set and DNA construction

Every model receives the same 100 synthetic classical-Chinese prompts from . The prompt construction and random projection seed remain fixed at 42 for every model, setting, and repeat.

For the original representation, each response is encoded using `all-mpnet-base-v2`. The 100 response embeddings are concatenated and mapped to a 128-dimensional DNA vector by a seeded Gaussian random projection. The current maximum response length is 1024 generated tokens. The saved metadata records at least the model ID, dataset, prompt count, temperature, top- p , sampling flag, generation seed, maximum length, projection seed, encoder, and DNA dimension.

Because the sentence encoder is primarily English-oriented and truncates long inputs, the final study will additionally report response-length and encoder token-length distributions, the fraction exceeding the encoder limit, and a sensitivity analysis with a Chinese or multilingual response encoder.

Decoding design

Factorial grid

The deterministic condition is

$$T=0, \text{ sampling disabled }, p=1.0 (\text{recorded only}).$$

Top- p is inactive when sampling is disabled; consequently, $T=0$ appears once and is not repeated as three nominal top- p cells.

For stochastic decoding, the complete grid is

$$T \in \{0.2, 0.3, 0.5, 0.7\}, p \in \{0.8, 0.9, 1.0\}.$$

This gives 13 decoding conditions in total. Two generation rounds are planned for every condition, giving 26 run cells and

$$100 \text{ models} \times 13 \text{ conditions} \times 2 \text{ rounds} = 2,600$$

successful artifacts before retries.

Generation rounds and seeds

The response-sampling seeds are:

Generation-repeat assignment. The projection seed remains 42 in all rows.

The shared $T=0.5$ cells require a duplicate/independence audit because both existing rounds use seed 42. Historical sweep artifacts also did not record a separate response-generation seed. In the August 5 snapshot, 572 of 2,528 successful artifacts record an explicit generation seed and 1,956 do not. These limitations affect causal temperature interpretation but do not invalidate paired method comparisons when every method reads exactly the same responses.

No response, DNA vector, or round may appear in both the training/reference split and the held-out query split. Generation seed and projection seed are stored separately so that response variability is not confounded with a change in the random projection.

Primary exact-model task

Let M be the fixed common model cohort. Given a query DNA vector $d_{m,s,r}$ generated by model m , decoding setting $s=(T, p)$, and held-out repeat $r=2$ in Table 3, each method evaluates every candidate $j \in M$. Let D_j^{train} denote the set of training DNA vectors available for candidate model j . Its gallery centroid is

$$\mu_j = \frac{1}{\sum_{x \in D_j^{train}} x}$$

The standard cosine similarity and cosine distance both take the query and the candidate centroid as inputs:

$$S_{cos}(d, \mu_j) = \frac{d^T \mu_j}{\|d\|_2 \|\mu_j\|_2}, D_{cos}(d, \mu_j) = 1 - S_{cos}(d, \mu_j).$$

Thus, the cosine rule may be written equivalently as

$$\hat{m}^{(cos)} = \arg \min_{j \in M} D_{cos}(d_{m,s,r}, \mu_j) = \arg \max_{j \in M} S_{cos}(d_{m,s,r}, \mu_j).$$

No square is applied in the standard cosine distance used by the evaluation code; each Euclidean norm already satisfies $\|d\|_2$, but the cosine value itself is not squared. The

squared norm appearing below belongs instead to the RBF kernel. For the SVM rules, the explicit class- j decision scores and predictions are

$$f_j^{(a)}(d) = \begin{cases} w_j^T z(d) + b_j, & a = \text{lin}, \\ \sum_{i=1}^{n_{sv}} \alpha_{ij} \exp(-\gamma \|z(d) - z(x_i)\|_2^2) + b_j, & a = \text{rbf}. \end{cases} \quad \hat{m}^{(a)} = \arg \max_{j \in M} f_j^{(a)}(d_{m,s,r}), a \in \{\text{lin}, \text{rbf}\}.$$

Here $z(\cdot)$ is the feature-wise standardization learned from the training split, (w_j, b_j) are the linear-SVM parameters, and $\{x_i, \alpha_{ij}\}_{i=1}^{n_{sv}}$ and γ are the RBF-SVM support vectors, effective class- j coefficients, and kernel bandwidth parameter, respectively. The prediction is correct only if $\hat{m}^{(a)} = m$. Thus, cosine and SVM methods are evaluated on identical multiclass labels, queries, and candidate sets.

Methods

Planned representation $\times$ classifier comparison.

Representation

Original 128-D DNA

Original 128-D DNA

Original 128-D DNA

Matched- K mean response embedding

Response distributions

Uncompressed RFFTrace

Compact DistDNA (projected RFFTrace)

The linear and RBF SVMs consume the full 128-dimensional DNA vector. The legacy linear SVM using only negative cosine distance is not considered an exact-model classification improvement because a monotone linear transformation of cosine distance cannot alter closed-set candidate ranking. It remains only a binary verification baseline.

Controlled comparison of old and new methods

Fairness is defined by holding the data and evaluation unit fixed while changing only the representation or decision rule. Every paired comparison uses the same model IDs, prompts, response files, decoding cell, reference/query split, candidate set, and metric.

Controlled method-comparison matrix.

Comparison	What changes	What remains fixed
Cosine vs linear/RBF SVM	Decision rule only	Same 128-D DNA, queries, labels, split, and candidate models
Mean cosine vs RBF-MMD at K	Representation /distance only	Same K responses per prompt, encoder, prompts, tokens, reference and query sets

Exact RBF-MMD vs RFFTrace	Kernel approximation only	Same embeddings, bandwidth, samples, prompts, reference and query distributions
RFFTrace vs compact DistDNA	Projection only	Same shared RFF map and data; one frozen Gaussian projection for every model and setting
$K=1$ point check	Kernel distance only	Same single response on each side; explicitly not evidence for a multi-sample advantage

Linear/RBF-SVM hyperparameters are selected only within repeat 1. The RBF-MMD bandwidth, shared random Fourier map, and optional compact projection are calibrated or sampled once using only reference/calibration data and then frozen. Method differences are computed per identical held-out query and summarized by a paired model-cluster bootstrap. Historical unknown seeds are shared by all methods in an exploratory comparison rather than being treated as a method difference.

Known-setting descriptive evaluation

For each held-out condition (T, p) and each method a , the experiment reports

$$A_a(T, p) = \frac{1}{|Q_{T,p}| \sum_{q \in Q_{T,p}} I[\hat{m}_{a,q} = m_q]}$$

Top-1 is primary; Top-3, Top-5, and mean reciprocal rank (MRR) are secondary. The change from the deterministic condition is

$$\Delta A_a(T, p) = A_a(T, p) - A_a(0).$$

All per-setting comparisons will use the same held-out model queries.

Unknown-temperature bounded deployment

The main deployment scenario assumes that the exact decoding parameters are not provided to the classifier. For a temperature bound

$$\tau \in \{0.2, 0.3, 0.5, 0.7\},$$

one classifier is trained using repeat 1 from every allowed setting with $T \leq \tau$. Temperature and top- p are omitted from its features. It is then evaluated on held-out repeat 2 from all allowed settings.

To avoid overweighting temperatures that happen to have more nominal cells, repeats are first averaged within each (T, p) cell. The primary mixed accuracy is the equal-cell mean

$$A_a^{mix}(\tau) = \frac{1}{\sum_{s \in S_\tau} A_a(s)}, S_\tau = \{(T, p) : T \leq \tau\}.$$

The robustness guarantee is summarized by the worst observed setting

$$A_a^{worst}(\tau) = \min_{s \in S_\tau} A_a(s).$$

These quantities directly answer how accurately the system performs when the decoding setting is hidden but its temperature is bounded.

Distributional and matched-budget evaluation

The formal method follows, where it is named RFFTrace and used as the proposed distributional DNA. For prompt q , model m , and decoding setting s , let $z_{m,s,q,r} \in R^d$ be the all-mpnet-base-v2 embedding of independent response r . The RBF kernel and its shared random Fourier feature map are

$$k_\sigma(a, b) = \exp\left(-\frac{\|a - b\|_2^2}{2\sigma^2}\right), \psi_\sigma(z) = \sqrt{\frac{2}{D}}$$

where $\omega_j \sim N(0, \sigma^{-2} I_d)$ and $b_j \sim Uniform(0, 2\pi)$. A single draw of all $\{\omega_j, b_j\}$ is shared across every model, prompt, setting, and split. The empirical distribution embedding for one prompt, followed by the complete uncompressed RFFTrace, is

$$\hat{\mu}_{m,s,q} = \frac{1}{K} \sum_{r=1}^K \psi_\sigma(z_{m,s,q,r}), \hat{v}_{m,s} = \frac{1}{\sqrt{t}} [\hat{\mu}_{m,s,1}; \dots; \hat{\mu}_{m,s,t}] \in R^{tD}.$$

Consequently,

$$\|\hat{v}_a - \hat{v}_b\|_2^2 \approx \frac{1}{t} \sum_{q=1}^t \widehat{MMD}_\sigma^2(\{z_{a,q,r}\}_{r=1}^K, \{z_{b,q,r}\}_{r=1}^K),$$

where exact MMD uses the biased/V-statistic estimator, including diagonal terms. The compact DistDNA vector is $\hat{t}_{m,s} = A \hat{v}_{m,s}$, with one fixed Gaussian projection $A_{ij} \sim N(0, 1/L)$. This vector can be used directly for Euclidean retrieval or as input to a linear/RBF SVM.

The RBF bandwidth is selected by the median pairwise-distance heuristic using only a disjoint subset of reference prompts. The calibration prompts, evaluation prompts, encoder, bandwidth, RFF draw, projection, and all seeds are then frozen. The implementation in provides exact RBF-MMD, the shared RFF map, prompt-wise kernel means, $1/\sqrt{t}$ concatenation, and compact Gaussian projection. The executable uses the real MPNet encoder and exports exact-MMD, RFFTrace, compact-DistDNA, single cosine, and matched- K mean-cosine predictions on identical data. The earlier is now explicitly a legacy hashing pilot; it is not the formal proposed method. The separate consumes exported vectors with explicitly disjoint train/test keys and enforces at least two independent

training vectors per model by default. It implements the downstream linear and RBF SVM probe, but does not remove the data requirement for valid tuning.

The planned sample-budget ablation is $K \in \{1, 2, 4, 8\}$, subject to available independent responses. For each K , the report will include Top-1, MRR, generated tokens, reference storage, and analysis time. Single cosine always uses one response; the primary fairness comparison at each K is matched- K mean cosine versus exact MMD, RFFTrace, and compact DistDNA. Exact MMD versus RFFTrace isolates RFF approximation error, while RFFTrace versus compact DistDNA isolates projection loss.

With the present two-round grid, only a disjoint $K=1$ reference/query comparison is immediately possible. At $K=1$, exact RBF-MMD is a kernel point-distance check and cannot demonstrate a multi-sample distributional advantage. A valid $K=2$ experiment needs four independent response sets per prompt—for example, reference rounds 1 and 3 and query rounds 2 and 4—with recorded, distinct generation seeds. The same-setting evaluation measures repeat stability; cross-setting evaluation freezes the reference setting and varies the query setting to measure decoding robustness.

The proposed method will be declared better than the strongest matched-budget baseline only if the paired Top-1 improvement has a model-cluster bootstrap 95% confidence interval whose lower endpoint is above zero on held-out data. MRR, worst-setting accuracy, compute, and storage are secondary checks. A compact DistDNA+SVM comparison additionally requires enough independent training vectors per class for nested hyperparameter selection; until those vectors exist, Euclidean retrieval is the primary formal test and the SVM result remains planned rather than implied.

Secondary verification and tree-aware analysis

Binary verification

If binary pairwise verification is reported, its tasks are separated into:

1. exact same-model versus unrelated;
2. lineage-only (off-diagonal related) versus unrelated; and
3. the legacy same-or-related versus unrelated task.

Training may downsample negative pairs or use class weighting. Testing uses all available pairs at their natural prevalence. The main binary metrics are precision–recall AUC, precision, recall, specificity, Matthews correlation coefficient, false positives per query, and ROC-AUC. Balanced-test accuracy may be shown only as an explicitly labelled diagnostic.

Model-tree distance

The Hugging Face relation exporter stores direct base-model edges in addition to connected components. Using the undirected direct-edge graph, the shortest tree distance is

$$d_{tree}(i, j) = \begin{cases} 0, & i = j, \\ 1, & i \text{ and } j \text{ share a direct edge,} \\ 2, 3, \dots, & \text{connected by the corresponding number of edges,} \\ \infty, & \text{disconnected.} \end{cases}$$

The final report will show DNA-distance distributions for exact, direct one-hop, indirect two-hop, indirect three-or-more-hop, same-component with unknown hop distance, and disconnected pairs. For Top-1 predictions, cumulative tree-aware accuracy will be reported as exact, within one hop, within two hops, and within the same connected component. This analysis describes error severity and does not replace exact-model accuracy as the primary endpoint.

Statistical analysis plan

Uncertainty

Every accuracy and MRR estimate will include a 95% confidence interval. Resampling will preserve dependence among observations:

- model IDs are resampled as clusters across methods and settings;
- tree-related analyses use model-tree component clusters where relevant.

Pair-level observations sharing a query model, reference model, or tree component will not be treated as independent Bernoulli trials. With fixed training and test roles for rounds 1 and 2, the primary confidence intervals quantify model-level rather than generation-repeat uncertainty.

Paired comparisons

Method differences are computed on identical held-out queries. The principal quantity is

$$\delta_{a,b}(T, p) = A_a(T, p) - A_b(T, p),$$

with a paired cluster-bootstrap confidence interval. The analysis will also test whether a method changes the temperature degradation slope, for example with a mixed-effects logistic model of the form

$$\text{logit Pr}(\text{correct}) = \beta_0 + \beta_1 T + \beta_2 p + \beta_3 \text{method} + \beta_4 (T \times \text{method}) + \beta_5 (p \times \text{method}) + u_{\text{model}} + u_{\text{repeat}}.$$

The interaction terms, rather than isolated cell differences, address whether a method mitigates robustness loss.

Multiplicity and exploratory analyses

Top-1 accuracy and bounded mixed accuracy are primary endpoints. Top-3, Top-5, MRR, tree-aware accuracy, encoder ablations, and projection ablations are secondary. Any unplanned subgroup or hyperparameter analysis will be labelled exploratory. If many

method-by-setting significance tests are presented, a false-discovery-rate correction will be applied.

Ablations and validity checks

The following checks are planned where artifacts and compute permit:

1. projection dimensions $\{64, 128, 256, 512\}$ and at least five projection seeds;
2. an unprojected or prompt-level distance baseline;
3. prompt bootstrap and split-half reliability;
4. response encoder comparison including a Chinese or multilingual encoder;
5. response-token and encoder-token length distributions and truncation rates by setting;
6. fixed $T=0$ gallery, same-setting repeat gallery, and multi-setting centroid gallery;
7. leave-one-setting-out evaluation for unseen decoding configurations;
8. performance by model size, architecture, and chat/base status; and
9. failure probability by model attributes and decoding setting.

Small-scale demonstration report using existing artifacts

Scope and split

To check that the analyses in Sections 6–11 are executable before the primary sweep is complete, small exploratory demonstrations were run on July 23, 2026, using only artifacts already present under . No new model responses were generated. Repeat 1 was used for training/reference data and repeat 2 for testing. Six complete stochastic cells were available:

$$T \in \{0.2, 0.3\}, p \in \{0.8, 0.9, 1.0\}.$$

The $T=0.5$ and $T=0.7$ cells were excluded because they did not have a sufficiently complete independent-repeat split. These demonstrations use convenience cohorts determined by artifact availability, not the fixed 100-model primary cohort. They are therefore pipeline checks and exploratory effect estimates only.

Executive summary

Table 6 maps each planned protocol component to what could actually be demonstrated. “Partial” means that the code path ran but the available cohort or repeat structure could not exercise the full planned analysis.

Coverage of Sections 6–11 by the small-scale demonstration.

Protocol section	Demonstrated analysis	Available scale	Status
6: Exact task	Cosine centroid, linear SVM, and RBF SVM on identical queries	Six cells; 84 common models	Complete demo
7: Bounded deployment	Equal-cell mixed and worst-setting accuracy for $\tau=0.2, 0.3$	89 and 84 models	Complete demo
8: Distributional	Single cosine, mean cosine, and exact RBF-MMD	Eight models; 20 prompts; at most two repeats	Partial
9: Verification	Exact-only linear probe at natural pair prevalence	95 models; 9,025 pairs per direction	Complete demo
9: Tree-aware	Exact versus disconnected error and distance summaries	95 models; no related pairs	Partial
10: Statistics	Model-cluster CIs and paired method differences	One held-out repeat per cell	Partial
11: Validity	Fixed- $T=0$ versus pooled-gallery diagnostic	86 versus 84 models	Partial

Exact classification and bounded deployment demo

For $\tau=0.2$, the common cohort contained 89 models and the training pool contained the three repeat-1 cells at $T=0.2$. For $\tau=0.3$, requiring all six cells reduced the common cohort to 84 models. The classifier was not given temperature or top- p . Table 7 reports held-out repeat-2 Top-1 accuracy for the $\tau=0.3$ pooled training experiment.

Exploratory held-out Top-1 accuracy with 84 common models.

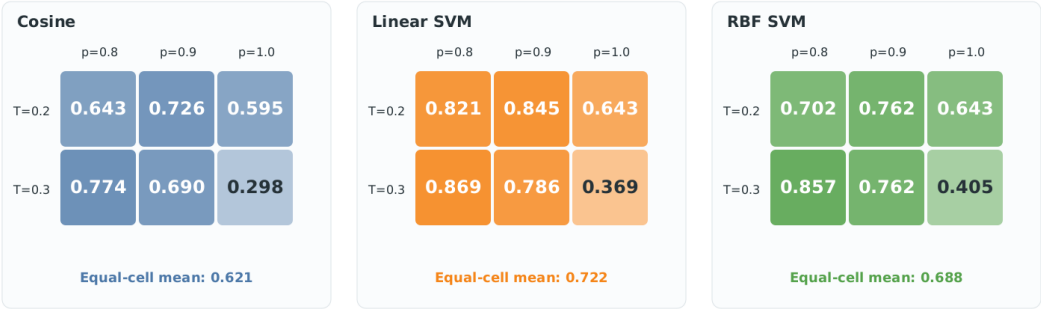
T
0.2
0.2
0.2
0.3
0.3
0.3

Figure 1 makes the setting-level pattern explicit. The linear SVM led in four cells and tied in one, while the RBF SVM was highest at $(0.3, 1.0)$. All methods dropped sharply in that cell,

so the demo does not support a general monotone temperature effect independent of top- p .

Held-out exact-model Top-1 by decoding setting

Repeat 1 pooled training; repeat 2 testing; 84 common models; temperature/top-p hidden from SVMs



Exploratory exact-model Top-1 by method and decoding cell. Darker cells indicate higher accuracy. The equal-cell means correspond to the $\tau=0.3$ pooled-training experiment.

The equal-cell bounded summaries are shown in Table 8. Confidence intervals resample model IDs as clusters across the included cells.

Exploratory bounded-deployment Top-1 summaries.

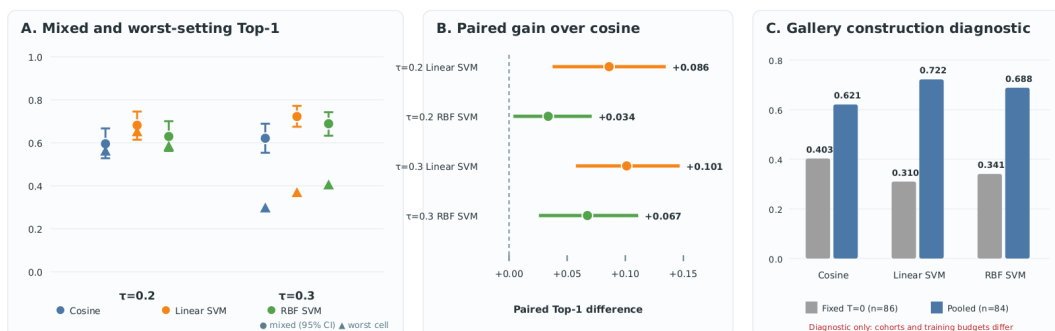
τ
0.2

On identical queries, the paired model-cluster bootstrap difference between linear SVM and cosine was 0.086 (95% CI 0.037 to 0.135) at $\tau=0.2$ and 0.101 (95% CI 0.058 to 0.147) at $\tau=0.3$. The corresponding RBF-SVM differences were 0.034 (95% CI 0.004 to 0.071) and 0.067 (95% CI 0.026 to 0.111). These intervals quantify only this availability-selected demo and do not test the primary hypotheses.

Figure 2 combines three views that should remain distinct in interpretation: bounded performance, paired within-query method gains, and the unpaired gallery diagnostic. In particular, the fixed-gallery bars are not a controlled estimate of gallery effects because the gallery cohort and training budget differ from the pooled condition.

Bounded deployment, paired gains, and gallery sensitivity

Exploratory summaries from the available-repeat demo



Bounded-deployment and sensitivity summary. Panel A shows equal-cell mixed accuracy with model-cluster 95% intervals and the worst observed cell. Panel B shows paired Top-1 differences relative to cosine. Panel C is an explicitly unpaired fixed-gallery versus pooled-gallery diagnostic.

Distributional and matched-budget demo

The exact RBF-MMD pilot ran successfully on eight lexicographically selected models, 20 common evaluation prompts, and two response repeats, using $T=0.2$, $p=0.8$ as the reference setting. Single-sample cosine, mean-cosine, and exact RBF-MMD each achieved Top-1 and MRR of 1.000 for the same-setting comparison and for cross-setting queries at $(0.3, 0.8)$ and $(0.3, 0.9)$. The same-setting split has only one repeat on each side, whereas the cross-setting comparisons pool both repeats per model on each side. This is a successful end-to-end demonstration, but the tiny cohort saturated all methods and therefore provides no evidence that the distributional representation is better. The available two repeats also do not support the planned $K \in \{1, 2, 4, 8\}$ ablation.

Distributional pilot Top-1 (MRR was also 1.000 in every entry).

Query comparison

- Same $(0.2, 0.8)$
- Cross $(0.2, 0.8) \rightarrow (0.3, 0.8)$
- Cross $(0.2, 0.8) \rightarrow (0.3, 0.9)$

This complete saturation is itself a useful demo diagnostic: a larger and more difficult cohort is required before representation differences can be estimated. It would be inappropriate to select a method from these pilot numbers.

Binary verification and tree-aware demo

An exact-only linear verification probe was trained on one $T=0.2$, $p=0.8$ repeat and tested on the other, in both directions, over 95 common models. Testing retained all 8,930 negative pairs per direction, giving a natural positive prevalence of $1/95$. Results are in Table 10.

Exploratory exact-only binary verification at natural prevalence.

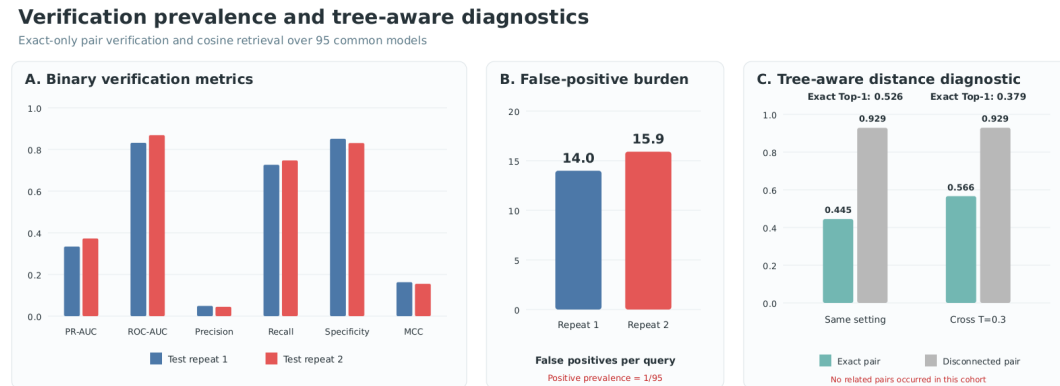
Test repeat

- 1
- 2

Despite the apparently strong ROC-AUC, the probe produced 1,329 and 1,511 false positives in the two directions, respectively. The resulting precision near 5% demonstrates why the primary verification report emphasizes PR-AUC, MCC, specificity, and false positives per query rather than balanced accuracy alone.

The tree-aware cosine demo used repeat 1 at (0.2, 0.8) as the gallery and 95 common models. Exact Top-1 was 0.526 for the same-setting repeat-2 query and 0.379 for the (0.3, 0.8) repeat-2 query. All 104 combined errors were between models disconnected in the exported relation graph, so within-one-hop, within-two-hop, and same-component accuracy did not exceed exact accuracy. Mean cosine distance was 0.445 versus 0.929 for exact and disconnected pairs in the same-setting comparison, and 0.566 versus 0.929 in the cross-setting comparison. No direct or indirect related pair occurred within this availability-selected cohort, so lineage-error severity could not be demonstrated.

Figure 3 emphasizes two practical limitations. First, ROC-AUC obscures the false-positive burden when only one of 95 candidate pairs is positive. Second, the tree-aware demo separates exact and disconnected pairs but cannot validate the intended one-hop and indirect strata because no such pair occurred in the common cohort.



Verification and tree-aware diagnostics. Panels A–B show binary metrics and false positives per query at natural prevalence. Panel C compares mean cosine distance for exact and disconnected pairs and annotates exact retrieval accuracy.

Validity-check demo and deferred items

As a limited gallery check, a fixed $T=0$ repeat-1 gallery was compared with the multi-setting pooled gallery on the same six query settings. Equal-cell mixed Top-1 with the fixed gallery was 0.403 for cosine, 0.310 for linear SVM, and 0.341 for RBF SVM over 86 common models. The corresponding pooled-gallery values were 0.621, 0.722, and 0.688 over 84

common models. Because the cohorts and number of training examples differ, this is a diagnostic rather than a paired method comparison; it merely confirms that gallery construction is consequential.

The remaining Section 10–11 analyses were deferred. Two temperatures and one held-out repeat are inadequate for the planned mixed-effects degradation model. Alternative projection dimensions/seeds, encoder comparisons, reliable encoder-token truncation counts, $K=4$ and $K=8$, lineage-only verification, and related-pair tree strata are not supported by the currently complete artifacts. Leave-one-setting-out and subgroup analyses are feasible later but were omitted to keep this demonstration deliberately small.

Demo conclusions and primary-study handoff

The demonstration supports the following limited conclusions:

- The exact-classification and bounded-deployment pipelines operate on disjoint generation repeats and produce aligned per-query scores for all three classifiers.
- Within this demo, the linear SVM had the highest mixed Top-1 at both temperature bounds, and its paired advantage over cosine was positive.
- Robustness was setting-dependent: $(0.3, 1.0)$ was substantially harder than the other five cells for every method.
- Natural-prevalence verification exposed a large operational false-positive burden that ROC-AUC alone would conceal.
- The distributional and tree-aware code paths run, but the present data are respectively too easy and too sparse in related pairs to compare the intended methods.
- Gallery construction materially changes the diagnostic accuracies, so the primary report must keep gallery definitions and training budgets fixed in paired comparisons.

Before a final primary claim is made, the study still requires recovery to at least the preregistered 80-model strict common cohort, a duplicate/independence audit of the two existing rounds, and the frozen paired evaluation on repeat 2. The current two-round design supports the exact-model cosine/SVM comparison. It does not yet support a multi-sample advantage claim: a fair $K=2$ test needs two independent reference responses and two independent query responses per prompt, while $K=4$ and $K=8$ require additional generation. The small-scale numbers remain exploratory until rerun on the frozen primary cohort.

Machine-readable outputs are stored under . The exact-classification, verification, distributional, fixed-gallery, and tree-aware subdirectories contain the selected cohorts, per-query predictions where applicable, aggregate CSV files, and JSON summaries. The three report figures are regenerated without plotting library dependencies using:

```
python scripts/plot_small_scale_section_6_11_demo.py
```


Execution and resource plan

The completed first backfill sequence used , the exact validated-100 cohort, and GPUs 0, 1, 2, and 4. It preserved successful historical artifacts and queued only missing or failed model-cell-round keys. The sequence ended on August 5 with 72 terminal failures. A failure-specific pass using was launched on the current A100 host on August 13 using GPUs 2–5. It writes to and preserves the original artifacts. The frozen counts in this report must be refreshed after it exits.

Twenty-four affected models lack only one slot. Restoring any 21 of these models across their missing cells is the minimum theoretical route from 59 to the preregistered 80-model strict common cohort. Full 100-model completion remains the target, but this prioritized recovery makes the analysis gate explicit.

The expected completion schedule below is measured from the August 5 status snapshot and assumes continued access to the existing GPUs. It separates the minimum analysis gate from optional multi-sample generation so that feasibility is visible rather than implied.

Expected completion schedule from the August 5 status snapshot.

Milestone	Expected time	Exit condition
Diagnose 72 terminal failures	1 working day	Failure-specific repair list
Targeted recovery and audit	2–3 working days	Strict common $N \geq 80$ and independent rounds
Frozen cosine/linear/RBF evaluation	1 working day	Paired tables and 95% model-cluster CIs
Report integration and review	1 working day	Primary report ready for supervisor review
Optional fair $K=2$ RBF-MMD generation	3–5 additional working days	Two independent samples on both sides

Therefore, the controlled cosine/SVM primary update is expected within five to six working days if recovery reaches $N \geq 80$. The multi-sample proposed-method claim has a separate, later completion date and is not placed on the critical path for the classifier comparison.

Transient failures are retried and recorded. Progress reports include planned, successful, failed, and missing models plus failure-status counts. Missing cells are never represented as zero accuracy and curves are not interpolated through absent cells.

Reproducibility and provenance

The experiment will preserve:

- exact cohort file and SHA-256 hash;

- prompt dataset and seed;
- model ID and available model metadata;
- temperature, top- p , sampling flag, generation seed, and token limit;
- response files and independent repeat suffixes;
- sentence encoder, DNA dimension, projection method, and projection seed;
- run manifests, status journals, failure summaries, predictions, and machine-readable metric tables; and
- the evaluation code and report version used to produce final figures.

For clarity, the August 5 completion snapshot has the following generation-seed provenance:

Generation-seed provenance among the 2,528 successful artifacts in the pre-recovery snapshot.

Provenance

Explicit generation seed recorded

Historical manifest without a separate recorded generation seed

Total artifact coverage

This table answers a counting issue directly: the 2,528 numerator does not exclude unknown-seed history. Any seed-clean analysis must construct and report its own denominator instead of reusing 2,600.

The unsuffixed historical gallery will not be used for the primary causal interpretation because its temperature and top- p were not recorded. It may be retained as a clearly labelled legacy reference-sensitivity analysis.

Planned results

Cohort and completion table

Frozen August 5 pre-recovery completion across both planned rounds.

Condition

$T=0$

$T=0.2$, all p

$T=0.3$, all p

$T=0.5$, all p

$T=0.7$, all p

Total

The strict common cohort across all 26 runs is 59. Setting-specific paired cohorts are reported in Table 2; therefore 97.2% slot completion must not be described as a 97-model complete-case cohort.

Per-setting exact classification

Available held-out exact-model Top-1. Starred entries are exploratory 84-model results, not final validated-100 estimates.

<i>T</i>
<i>T</i>
0.0
0.2
0.2
0.2
0.3
0.3
0.3
0.5
0.5
0.5
0.7
0.7
0.7

*Repeat 1 was used for pooled training/reference construction and repeat 2 for testing on the same 84 availability-selected models. The Proposed column remains pending: a same-query $K = 1$ RBF-MMD check is feasible, but a multi-sample claim requires additional matched responses.

Bounded unknown-temperature deployment

Temperature-agnostic deployment results. The classifier is not given temperature or top- p .

τ
0.2

Planned figures

The final report will contain:

1. exact Top-1 versus temperature, faceted by top- p , with one curve per representation/classifier and 95% confidence bands;
2. accuracy loss relative to $T=0$ and estimated degradation slopes;
3. mixed and worst-setting accuracy versus temperature bound τ ;
4. stacked tree-aware Top-1 error severity versus temperature; and
5. accuracy versus response-sample/token budget for the mean and distributional representations.

Planned interpretation

The final conclusions will be limited to what the completed protocol supports:

- the effect of query-side decoding conditions relative to explicit, independently generated galleries;
- classifier mitigation on the same exact-model task;
- representation gains under matched budgets; and
- expected and worst observed performance when decoding parameters are hidden but temperature is bounded.

No lineage-robustness conclusion will be based on aggregate positives dominated by same-model pairs. No method will be declared superior from a point estimate alone, and no causal temperature conclusion will use the historical unknown-setting gallery.

Execution commands

Build or verify the fixed cohort and refresh the live grid:

```
python3 scripts/build_validated100_config.py \
  --manifest out/top_p_control_t05_validated100_20260719/\
top_p_control_t05_t05_p06_r2/manifest.json \
  --output configs/rand_chinese_validated100_latest.jsonl
python3 scripts/build_primary_run_grid.py
```

The completed missing/failed backfill sequence was launched with:

```
env GPUS=0,1,2,4 MAX_CONCURRENT_GPUS=4 \
  bash scripts/resume_primary_robustness_grid.sh
```

After targeted recovery and the independence audit, run the exact-model and tree-aware analyses:

```
bash scripts/run_primary_analysis.sh
```

Run the implemented formal $K=1$ RFFTrace diagnostic after the common cohort reaches the minimum gate:

```
env ENCODER_DEVICE=cuda:0 bash scripts/run_primary_rfftrace_k1.sh
```

The same Python entry point supports a future valid $K=2$ experiment by passing `--reference-repeats 1,3 --query-repeats 2,4 --samples-per-side 2` after independent rounds 3 and 4 exist. It must not be run by reusing the same response twice.

The primary analysis outputs are written beneath . The exact-model evaluator produces per-setting metrics, per-query predictions, bounded-deployment tables, cohort metadata, and JSON summaries.

Implementation map

File	Purpose
File	Purpose
	Launch the 100-model, 13-condition, two-round extraction sweep.
	Resume missing and failed validated-100 grid cells without regenerating successful artifacts.
	Produce slot-completion and repeat-level cohort audits.

Construct the fixed train/test repeat split and run primary analyses.

Compare cosine centroids and multiclass SVMs on exact identity; produce bounded unknown-temperature results and confidence intervals.

Produce hierarchical retrieval and tree-distance error summaries.

Run legacy and separated binary verification tasks using all test negatives by default.

Legacy character-hashing pilot for single-sample, mean-cosine, and exact RBF-MMD retrieval; not the formal DistDNA implementation.

Implement exact biased RBF-MMD, shared random Fourier features, prompt-wise RFFTrace vectors, and optional compact Gaussian projection.

Run formal MPNet-based, calibration-separated, matched-budget exact-MMD, RFFTrace, and compact-DistDNA retrieval and export reusable vectors.

Freeze the primary-cohort $K = 1$ same/cross-setting diagnostic command.

Evaluate exported RFFTrace or compact-DistDNA vectors with held-out linear and RBF SVMs while enforcing disjoint vector keys.

Check kernel geometry, MMD/RFF convergence, scaling, projection, seeds, and package exports.

Preserve legacy pilot MMD, repeat-split, and all-record prompt-intersection regression checks.

Check response discovery, actual model IDs, common prompts, required repeats, and disjoint prompt splitting.

Check SVM split leakage rejection and held-out multiclass evaluation.

Export direct parent/base edges and connected components.

Report completion, missing models, and failure-status counts.